

Q_{AI} $(X \otimes X)^2 = II$ $(X \otimes X)(Y \otimes Y) = -II$ Entanglement detection using Bell (correlation) measurement

$$B_i = \frac{1}{\sqrt{2}} (|00\rangle + |11\rangle) \quad \begin{aligned} X \otimes |B_i\rangle &= |B_i\rangle \\ Y \otimes |B_i\rangle &= -|B_i\rangle \\ Z \otimes |B_i\rangle &= |B_i\rangle \end{aligned}$$

$X \otimes X$	$X \otimes Y$	$X \otimes Z$
$Y \otimes X$	$Y \otimes Y$	$Y \otimes Z$
$Z \otimes X$	$Z \otimes Y$	$Z \otimes Z$

$X \otimes X$	$X \otimes Y$	$X \otimes Z$
$Y \otimes X$	$Y \otimes Y$	$Y \otimes Z$
$Z \otimes X$	$Z \otimes Y$	$Z \otimes Z$

$$|B_i \otimes B_i\rangle = \frac{1}{4} [II + XX - YY + ZZ]$$

$$p_i = \text{tr}(\rho |B_i \otimes B_i\rangle) = \langle B_i \otimes B_i |$$

$$\frac{1}{4} [\underbrace{\text{tr}(\rho XX)}_{\langle XX \rangle} + \underbrace{\text{tr}(\rho YY)}_{\langle YY \rangle} - \underbrace{\text{tr}(\rho YY)}_{\langle YY \rangle} + \underbrace{\text{tr}(\rho ZZ)}_{\langle ZZ \rangle}]$$

$$0 \leq \frac{1}{4} [1 + \langle XX \rangle - \langle YY \rangle + \langle ZZ \rangle] \leq \frac{1}{2} < 1$$

$$0 \leq 1 + \langle XX \rangle - \langle YY \rangle + \langle ZZ \rangle \leq 2 < 4$$

$$-1 \leq \langle XX \rangle - \langle YY \rangle + \langle ZZ \rangle \leq 1 < 3$$

$$\begin{aligned} -1 &\leq +\langle A \rangle - \langle B \rangle - \langle C \rangle \leq 1 < 3 \leftarrow |B_2 \otimes B_2\rangle \\ -1 &\leq +\langle A \rangle - \langle B \rangle + \langle C \rangle \leq 1 < 3 \leftarrow |B_1 \otimes B_1\rangle \\ -1 &\leq -\langle A \rangle + \langle B \rangle + \langle C \rangle \leq 1 < 3 \leftarrow |B_3 \otimes B_3\rangle \\ -1 &\leq -\langle A \rangle - \langle B \rangle - \langle C \rangle \leq 1 < 3 \leftarrow |B_4 \otimes B_4\rangle \end{aligned}$$

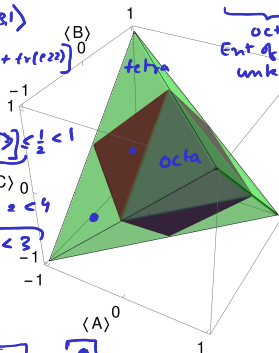
$$\rho = \frac{1}{4} \left(II + \sum_{k=1}^3 \langle \sigma_k \otimes I \rangle \sigma_k \otimes I + \sum_{l=1}^3 \langle I \otimes \sigma_l \rangle I \otimes \sigma_l + \sum_{k,l=1}^3 \underbrace{\langle \sigma_k \otimes \sigma_l \rangle}_{t_{kl}} \sigma_k \otimes \sigma_l \right)$$

$$[\sigma_k \otimes I, \rho] = \text{tr}(\sigma_k \otimes I \rho)$$

$$0 \leq p_i \leq \frac{1}{2} < 1$$

octa
Ent of ρ remains unknown

$$\sum_{i=1}^4 p_i = 1$$



$$H \otimes |B_i \otimes B_i\rangle H \otimes I = \frac{1}{4} [II + 2X + YY + XZ]$$

$$S \otimes I |B_i \otimes B_i\rangle S \otimes I = \frac{1}{4} [II + YX + XY + ZZ]$$

$$4^2 - 1 = 15$$



R. Horodecki and M. Horodecki, Phys. Rev. A **54**, 1838 (1996). O. Gühne and M. Lewenstein, Phys. Rev. A **70**, 022316 (2004). T.-C. Wei and P. M. Goldbart, Phys. Rev. A **68**, 042307 (2003). O. Gühne and G. Tóth, Physics Reports **474**, 1 (2009). A. Sehwat, arXiv:1611.09760 [quant-ph] (2017).

Q π_{AI} For a single spin $S_x = \frac{\hbar}{2} \sigma_x$ Total spin measurement $\hbar = 1$ Local $n=2$



Label for a spin-1/2 particle

$$S_x := \frac{1}{2} \sum_{i=1}^n \sigma_x^{(i)} = \frac{1}{2} (X \otimes I + I \otimes X) \quad \text{global}$$

$$\frac{1}{2} (XI + IX) \quad \frac{1}{2} (XI + IX) = \frac{1}{4} [II + XX + YY + ZZ]$$

$$(S_x)^2 := \frac{1}{4} \sum_{i,j=1}^n \sigma_x^{(i)} \sigma_x^{(j)} = \frac{1}{2} (I \otimes I + X \otimes X)$$

$$|B_{1,3}\rangle = \frac{1}{\sqrt{2}} (|00\rangle \pm |11\rangle)$$

$$|B_{2,4}\rangle = \frac{1}{\sqrt{2}} (|01\rangle \pm |10\rangle)$$

$$S = (S_x)^2 + (S_y)^2 + (S_z)^2 = \frac{1}{2} (3II + XX + YY + ZZ)$$

$$= \vec{S} \cdot \vec{S}, \quad \vec{S} = (S_x, S_y, S_z)$$

$$= \begin{bmatrix} 2 & 0 & 0 & 0 \\ 0 & 1 & 1 & 0 \\ 0 & 1 & 1 & 0 \\ 0 & 0 & 0 & 2 \end{bmatrix} \begin{bmatrix} 0 \\ 1 \\ 1 \\ 0 \end{bmatrix} \frac{1}{\sqrt{2}}$$

$$S_z = \frac{1}{2} (ZI + IZ) = \begin{bmatrix} 1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & -1 & 0 \\ 0 & 0 & 0 & -1 \end{bmatrix} \begin{bmatrix} 0 \\ 1 \\ 1 \\ 0 \end{bmatrix} \frac{1}{\sqrt{2}} = |B_{2,4}\rangle$$

ortho
H₂

$S_z 00\rangle = 00\rangle$	$S 00\rangle = 2 00\rangle$
$S_z 11\rangle = - 11\rangle$	$S 11\rangle = 2 11\rangle$
$S_z B_2\rangle = 0 B_2\rangle$	$S B_2\rangle = 2 B_2\rangle$

triplet
 $\{|00\rangle, |11\rangle, |B_2\rangle, |B_4\rangle\}$

$ 00\rangle$	$\uparrow\uparrow$	$\frac{1}{2} + \frac{1}{2} = 1$
$ 01\rangle$	$\uparrow\downarrow$	$\frac{1}{2} - \frac{1}{2} = 0$
$ 10\rangle$	$\downarrow\uparrow$	$-\frac{1}{2} + \frac{1}{2} = 0$
$ 11\rangle$	$\downarrow\downarrow$	$-\frac{1}{2} - \frac{1}{2} = -1$

$$S_z|01\rangle = \frac{1}{2} ZI|01\rangle + \frac{1}{2} IZ|01\rangle = \left(\frac{1}{2} - \frac{1}{2}\right)|01\rangle$$

para

$S_z B_4\rangle = 0 B_4\rangle$	$S B_4\rangle = 0 B_4\rangle$
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$$XX = \begin{bmatrix} 1 & 0 & 0 & 0 \\ 0 & -1 & 0 & 0 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & -1 \end{bmatrix} \quad YY = \begin{bmatrix} 0 & -i & 0 & 0 \\ i & 0 & 0 & 0 \\ 0 & 0 & 0 & -i \\ 0 & 0 & i & 0 \end{bmatrix} \quad ZZ = \begin{bmatrix} 1 & 0 & 0 & 0 \\ 0 & -1 & 0 & 0 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & -1 \end{bmatrix} \quad II = \begin{bmatrix} 1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 \end{bmatrix}$$

